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RESEARCH ARTICLE

Super-Wideband BAVA With Enhanced Radiation Characteristics Using an Integrated Biconvex Magneto-Electric Lens for 5G mmWave Communications

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ABSTRACT A super-wideband balanced antipodal Vivaldi antenna (BAVA) is proposed incorporating two pairs of lenses integrated within the tapered slot region. The first pair consists of biconvex magneto-electric lenses (ME-L), oriented horizontally with respect to the slot axis, while the second pair comprise biconvex electric lenses (E-L), positioned orthogonally to the ME-L. Through a controlled field shaping, the ME-L pair enables efficient field coupling to enhance mode symmetry across the tapered arms and provides a smoother impedance transition to free space. In contrast, the E-L pair introduces predominantly electric-field coupling. Both lens pairs act as phase-correcting directors that redistribute the aperture's electric field, resulting in a more uniform phase front, increased gain, and improved beam confinement. In addition, several open-ended elliptical slits are incorporated within the tapered arms to enhance broadband impedance matching at the lower end of the operating band. The antenna exhibits an excellent reflection coefficient across the 5–40 GHz frequency range and demonstrates a directive end-fire radiation pattern, with a maximum measured gain of 14.5 dBi within the 20–30 GHz band. Furthermore, stable radiation characteristics are observed, with a beam squint less than 2° over the 10–40 GHz range. The antenna features a compact, low-profile, and fully planar structure. Unlike conventional dielectric lenses, which are often sensitive to alignment and integration tolerances, the proposed ME-L and E-L lenses enable seamless integration with the antenna structure and support automated fabrication processes.

INDEX TERMS 5G-mmWave communication, AVA, electric and magneto-electric lens, super wideband antenna.

I. INTRODUCTION

Microwave and millimeter-wave technologies form the backbone of modern communication systems, enabling significant advancements and supporting a wide spectrum of emerging applications [1]. Each frequency range offers distinct advantages that determine its role and versatility in various systems. Microwave frequencies are extensively used in long-range telecommunications, satellite links, radar systems, and wireless networks due to their efficient propagation

characteristics and well-established infrastructure [2]. In contrast, millimeter-wave frequencies are increasingly employed in applications that demand high data rates and precision, such as 5G mobile networks, point-to-point links, and high-resolution imaging systems [3]. To address the diverse requirements of advanced communication platforms, antenna designs capable of operating across both microwave and millimeter-wave bands have attracted growing attention. Among these, the antipodal Vivaldi antenna (AVA) has emerged as a promising wideband radiator, widely adopted in satellite communication, radar, electronic warfare, remote sensing, and microwave imaging due to its broad bandwidth,

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high directivity, and ease of fabrication [4]. The AVA offers several advantages, including low weight, compactness, low manufacturing cost, wide scan angle capability, and straightforward system integration. Nonetheless, many modern applications now demand even higher gain and more stable directivity across an extended frequency range to meet stringent performance requirements, such as improved range resolution and imaging accuracy [5]. Despite its merits, the conventional Vivaldi antenna exhibits limitations at higher frequencies, where performance degradation manifests as reduced directivity, gain loss, and radiation pattern distortion. These effects often arise from unwanted surface currents along the taper termination and phase reversal phenomena; both linked to the antenna's inherent structural constraints. Furthermore, asymmetry between the ground and feed flares introduces field imbalance, leading to elevated cross-polarization levels and beam squint. Consequently, enhancing high-frequency performance and ensuring stable radiation characteristics remain key challenges in advancing the Vivaldi antenna design [6].

In recent years, extensive research has been devoted to enhancing the radiation performance of antipodal Vivaldi antennas for wideband and millimeter-wave applications. Various structural modifications have been proposed to improve gain, beam stability, sidelobe suppression, and polarization purity. Razib et al. [7] and Moosazadeh and Kharkovsky [8] incorporated dielectric lenses—half-spherical and trapezoidal respectively, to improve directivity, beam focusing, and high-frequency gain. Similarly, Karmakar et al. [9] employed dual Koch fractal lenses, while Shi et al. integrated a metamaterial array and elliptical slot director to achieve broad impedance bandwidth and stable radiation across multiple bands [10]. These designs achieved high gain, low sidelobe levels, and reduced beam squint; however, their multilayer or dielectric lens-based configurations inherently increased fabrication and alignment complexity, aperture size, and sensitivity to dielectric placement and tolerances.

To overcome the structural and fabrication challenges of multilayer or dielectric lens-loaded AVAs, several simplified yet effective approaches have been introduced. Moosazadeh [1] achieved low sidelobes and high gain using a tapered Teflon dielectric loading (AVA-STD) without resorting to metamaterials or complex stacking, while Honari et al. enhanced bandwidth and miniaturization through dual exponential strip arms in a single-layer configuration, maintaining good gain and polarization stability [11]. Meanwhile, designs by Bang et al. [5] and Harel et al. [12] improved high-frequency radiation performance using parasitic patches and Chebyshev-tapered flares respectively, both offering effective suppression of pattern distortion and cross-polarization without increasing structural complexity [12]. Golla and Nelaturi further extended these concepts to array configurations, integrating parasitic and corrugated features to boost directivity and front-to-back ratio at 28 GHz,

though high-frequency dispersion and minor gain degradation remained [13]. Recent studies highlight a trade-off between performance enhancement and structural simplicity. Slot- and slit-based optimization techniques improve gain and return loss without increasing antenna size; however, they require precise parametric tuning and are sensitive to fabrication tolerances [14]. In contrast, metamaterial-based approaches enhance gain and directivity but often increase antenna profile, size, and integration complexity, limiting their suitability for compact systems [15].

Collectively, these studies demonstrate that while the inclusion of dielectric, fractal, and metamaterial elements significantly enhances AVA radiation characteristics, such gains often come at the expense of fabrication simplicity and efficiency consistency. In contrast, recent single-layer and taper-optimized designs achieve comparable radiation stability with reduced complexity, positioning them as promising solutions for compact, wideband, and high-frequency systems.

In this article, we propose an AVA augmented with biconvex magneto-electric lenses and electrical parasitic lenses for 5G millimetre-wave communications. Introducing a biconvex ME-L and E-L elements between the antipodal flares enhances radiation through coordinated field coupling, mode balancing, and impedance control. These elements couple simultaneously to the electric and magnetic fields in the tapered-slot region, reshaping the local field distribution and acting as a focusing lenses that concentrate surface current and electric field near the aperture. This increases the effective radiating current density and strengthens end-fire directivity. In addition, the interaction of orthogonal electric and magnetic responses establishes an ME resonance that improves impedance matching and reduces reflection ($|S_{11}|$) over a wide bandwidth. At higher frequencies, where asymmetric surface-current phase on the upper and lower flares can induce beam squint and cross-polarization, the ME lenses supply balanced reactive coupling to compensate phase delay and maintain in-phase end-fire radiation. The proposed antenna demonstrates excellent radiation performance, achieving a high and stable realised gain of 14.5 dBi over a wide millimetre-wave frequency range from 20 GHz to 40 GHz, while maintaining a beam squint of less than 2°. Moreover, the proposed design eliminates the need for external dielectric lenses to obtain high-gain characteristics, as commonly reported in the literature. Such lens-based approaches often introduce several drawbacks, including increased overall antenna size, enlarged substrate footprint, and the requirement for multilayer configurations. In addition, dielectric lenses are frequently fabricated from materials different from the antenna substrate, which complicates integration and limits compatibility with automated machine-based fabrication processes.

The remainder of this paper is organized as follows: Section II describes the antenna geometry and explains the operating role of the parasitic lenses. Section III presents the

simulated results, while Section IV outlines the design, fabrication, and measured results that validate the performance improvements. Finally, Section V concludes the paper.

II. ANTENNA GEOMETRY

The geometry of the proposed AVA is illustrated in Fig. 1. The antenna is fabricated on a Rogers RO4003C substrate with a thickness of 0.506 mm, a dielectric constant (ϵ_r) of 3.55, and a loss tangent of 0.0027 at the centre frequency of 18 GHz. The overall dimensions of the antenna are 64.2 mm $\times$ 40.3 mm ($L \times W$). The AVA comprises three main sections: a pair of exponentially tapered radiators, a microstrip feed line of 1.05 mm width providing a characteristic impedance of 50 Ω , and an elliptically tapered ground plane printed on the opposite side of the substrate. Furthermore, the proposed design integrates two distinct configurations of biconvex lenses within the tapered slot region—namely, biconvex magneto-electric lenses oriented horizontally and sandwich-like biconvex electrical lenses oriented vertically to enhance radiation performance across a wide frequency range, particularly around the 28 GHz millimeter-wave band.

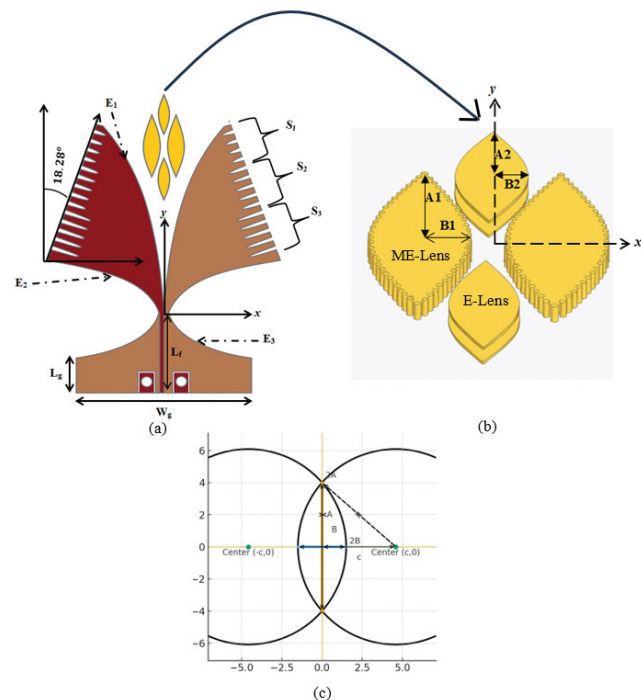


FIGURE 1. Geometry of proposed balance antipodal Vivaldi antenna (BAVA); (a) Overall structure, (b) ME and E-Lenses structure, (c) biconvex lens diagram construction.

In addition, several open-ended elliptical slits are incorporated into the antenna structure to improve impedance matching at lower frequencies. The antenna design and its evolution toward the final configuration, as depicted Fig. 1, were carried out through three principal stages as follows:

The design process begins with the development of the upper metallic flare, which is shaped using an exponentially

tapered profile to efficiently radiate the traveling wave and achieve excellent impedance matching across an ultra-wide frequency band. The exponential curves E_1 and E_2 that define the slot contour are expressed by the following equations [16]:

$$E_1 = 1.54 - e^{nx}, \quad 0 \leq x \leq 36, \quad n = 0.069 \quad (1)$$

$$E_2 = 0.5 - e^{mx}, \quad 0 \leq x \leq 8.98, \quad m = 0.334 \quad (2)$$

The ends of these exponential curves are joined by a straight line inclined at an angle of 18.28° from the vertical axis, causing the aperture (or termination) of the metallic flare to form the same angle with the vertical axis, as illustrated in Fig. 1 (a). Regarding the brown-colored ground plane, the upper flare represents the mirror image of the ground flare, whereas the ground flare consists of a base with a width W_g and a height L_g , terminating at the top with an exponential curve (E_3), which is defined by the following equation [16].

$$E_3 = -0.5 + e^{kt}, \quad 0 \leq t = -y \leq 8.98, \quad k = 0.334 \quad (3)$$

In the subsequent design stage, a set of open-ended elliptical slits was incorporated along the radiating flares of the AVA, as illustrated in Fig. 1 (a), to improve impedance matching over a wide frequency range, particularly at the lower end of the band. These slits were organized into three groups, S_1 , S_2 , and S_3 , each containing five elements with progressively increasing lengths. The semi-major axes assigned to groups S_1 , S_2 , and S_3 were 2 mm, 3 mm, and 5 mm, respectively, while all slits shared a common semi-minor axis of 0.3 mm. It is presumed that these slits act as reactive loading features that may influence the local surface current distribution and introduce additional resonant effects [17]. Each group of slits appears to contribute to a gradual impedance transformation along the flare, where the shorter slits (S_1) predominantly affect the higher-frequency region by providing subtle capacitive loading near the aperture. In contrast, the longer slits (S_3) could introduce stronger inductive effects, thereby improving matching at the lower-frequency end as shown in Fig. 2. Collectively, this configuration facilitates a multi-resonant response that broadens the operational bandwidth and reduces the $|S_{11}|$. As suggested by the simulated surface current distribution of Fig. 2, concentrated currents are observed around the slit edges, which indicate enhanced electric and magnetic coupling. Such coupling could extend the effective current path and smooth the impedance transition between the feed and the free space region, leading to an improvement in the overall impedance matching and radiation parameters across the working band.

In the third step, two types of biconvex lenses, known as magneto-electrical lenses and electrical lenses, were integrated in the tapered slot region between upper flare and lower one. These lenses were designed in different sizes, each formed by the intersection of two circles with identical radii. The first pair ME-L is horizontally aligned along the x-axis and has a larger dimension, while the second pair E-L is

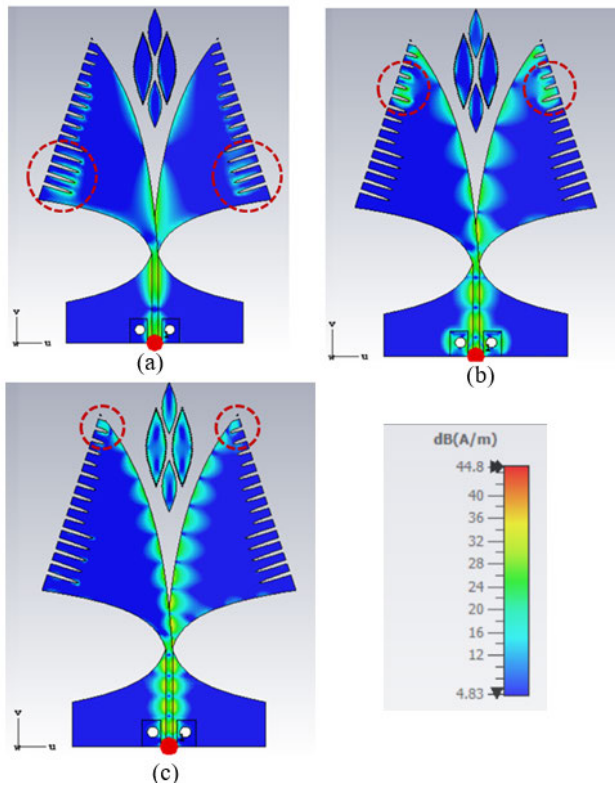


FIGURE 2. Surface current distribution around the open-ended elliptical slits at different frequencies: (a) 8 GHz, (b) 18 GHz, and (c) 28 GHz.

vertically aligned along the y-axis, being relatively smaller. In the design illustrated in Fig. 1 (b), the horizontal pair is generated by two circles with centers located at $(0, \pm c)$, where $c = 4.58$ mm, and each circle has a radius of $R = 6.08$ mm. Accordingly, the boundary arc of the biconvex shape of Fig. 1 (c) can be expressed as follows [18]:

$$(x + c)^2 + y^2 = R^2, \quad (x - c)^2 + y^2 = R^2 \quad (4)$$

Semi-length (center to tip),

$$A_1 = \sqrt{R^2 - c^2} \quad (5)$$

Semi-width (at midline),

$$B_1 = R - c \quad (6)$$

Therefore, the parameters A_1 and B_1 for the ME-L are 6 mm and 1.5 mm, respectively. Similarly, the semi-length and semi-width of the E-L are 4 mm and 1.1 mm, respectively.

The ME-L is realized by vertically connecting a pair of identical biconvex lenses positioned on the top and bottom surfaces of the substrate, respectively. These lenses are precisely aligned and electrically connected through 60 metallized vias, enabling the structure to operate analogously to a magnetic dipole radiator and thereby forming an effective magneto-electric lens configuration [19]. This arrangement facilitates the simultaneous excitation of electric and magnetic field sources: the electric field across the antipodal flares is concentrated within the narrow gap

between the upper and lower lens halves, while the magnetic field circulates through the vias, forming inductive current loops, as illustrated in Fig. 3 (a) and (c) respectively. The resulting dual-mode coupling focuses the near-field distribution toward the antenna aperture, thereby enhancing the effective aperture efficiency and suppressing undesired lateral field leakage, as shown in Fig. 4. Consequently, improvements in the antenna's realised gain and directivity are achieved.

The E-L pair, oriented along the y-axis and fabricated without vias, is introduced to provide a predominantly electric coupling mechanism. Despite their relatively smaller dimensions, these lenses function as phase-correcting directors that redistribute the electric field across the antenna aperture, as illustrated in Fig. 4. This redistribution promotes a more uniform phase front and reinforces end-fire radiation. Consequently, additional gain enhancement and beam narrowing are achieved, particularly in the elevation plane (θ -Plane).

The combined implementation of both ME-L and E-L forms a hybrid focusing structure that effectively narrows the main beam in both the ϕ and θ -planes. The ME-L lenses primarily suppress sidelobes and compress the beamwidth in the ϕ -plane by inducing constructive interference between the surface currents along the flare edges and the metallic lens boundaries. In contrast, the E-L lenses improve radiation symmetry in the θ -plane, leading to a squeeze half-power beamwidth (HPBW) and an enhanced front-to-back ratio. One of the most notable improvements achieved, in addition to gain enhancement, is the mitigation of beam squint, which commonly arises in wideband tapered slot antennas due to frequency-dependent phase velocity variations along the exponential flares. The ME-L and E-L compensates for these variations by balancing the electromagnetic energy propagation through both electric and magnetic coupling channels. This mechanism maintains a more uniform phase distribution across the aperture, ensuring frequency-independent radiation patterns and reduced main-beam deviation across the operating band. At the same time, the distributed inductive and capacitive loading created by the vias and the biconvex lens boundaries act as an effective impedance transformer which improve the return loss response as well. The metallic vias contribute inductive reactance, while the biconvex boundaries introduce capacitive coupling between the flare arms. These reactive effects jointly generate a gradual impedance transition between the microstrip feed and the radiating aperture, leading to broadband impedance matching and lower $|S_{11}|$.

III. SIMULATION RESULTS AND PERFORMANCE IMPROVEMENTS

This section presents a comprehensive analysis of the simulated results obtained for the proposed AVA configurations incorporating dual pairs of biconvex ME-L and E-L lenses combined with flare-integrated slits, as described in the preceding section. The investigation encompasses a comparative

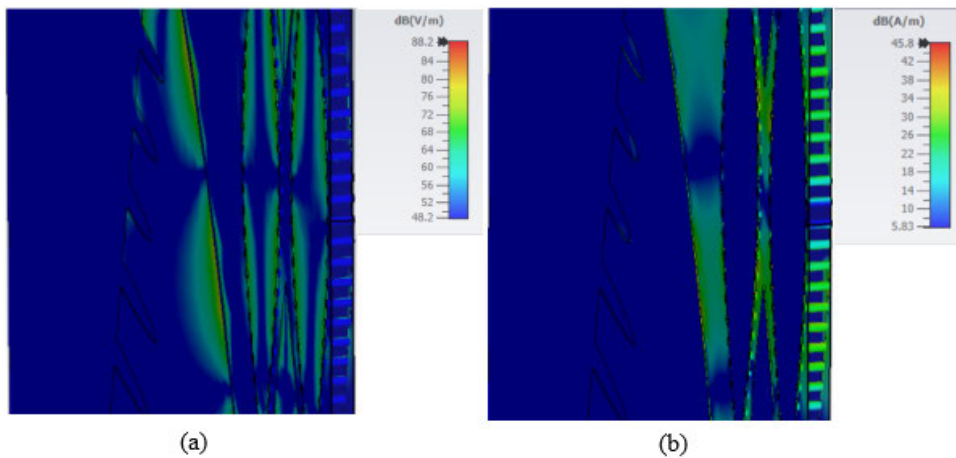


FIGURE 3. Electromagnetic field distributions around the magneto-electric biconvex lenses: (a) electric-field distribution around the biconvex lenses; (b) magnetic-field distribution around the vias.

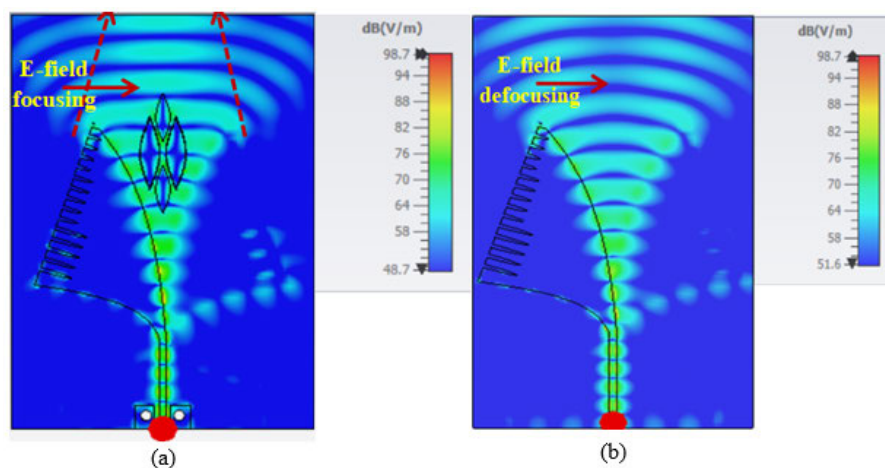


FIGURE 4. Electric-field distributions of the proposed BAVA antenna: (a) with integrated magneto-electric and electric lenses; (b) without lenses.

evaluation of several key performance parameters, including the reflection coefficient, realized gain, radiation efficiency, radiation pattern, beam squint, side-lobe level (SLL), front-to-back ratio, HPBW, surface current distribution, fabrication tolerance analysis, and overall radiation characteristics.

A. INPUT IMPEDANCE

The simulated reflection coefficients of the proposed AVA at various design stages are illustrated in Fig. 5, covering the frequency range from 10 GHz to 40 GHz, which is suitable for a wide range of microwave and millimeter-wave applications, particularly 5G mmWave communications. Notably, all four simulated responses exhibit return loss values below -10 dB, indicating satisfactory impedance matching across the operating band. From the figure, it can also be observed that the inclusion of the slits significantly improved the antenna's return loss performance by, reducing it by approximately 10 dB or more compared with the reference antenna,

particularly at higher frequencies within the 20–40 GHz range. The addition of the ME-L lenses did not produce a noticeable change in the reflection coefficient, as it remained nearly the same as in the case with slits only. In contrast, incorporating the E-L lenses caused a slight variation in the return loss; however, the performance remained stable, maintaining S_{11} below -10 dB across the operating band.

B. ANTENNA GAIN

Fig. 6 illustrates the simulated gain performance of the proposed AVA antenna across the entire frequency spectrum (10–40 GHz). The antenna demonstrates a nearly stable response across this range, achieving a peak simulated gain of 14.5 dBi. Notably, the gain remains consistently above 13 dB within the 20–30 GHz band, signifying a marked enhancement around the 5G millimetre-wave region (≈ 28 GHz). As shown in the figure, each successive modification stage of the reference antenna results in a

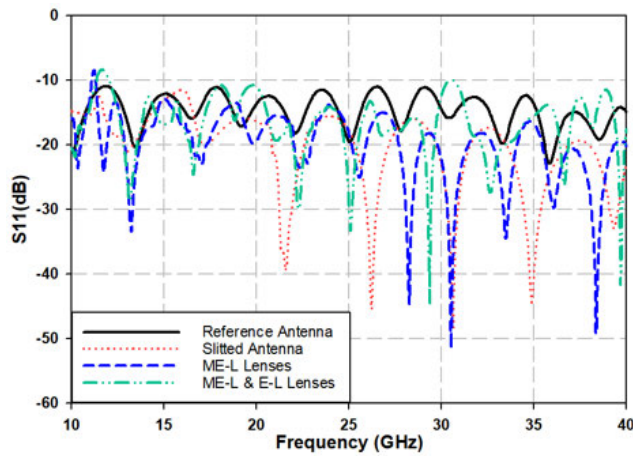


FIGURE 5. Reflection coefficient (S_{11}) comparison across different design stages of the proposed BAVA antenna.

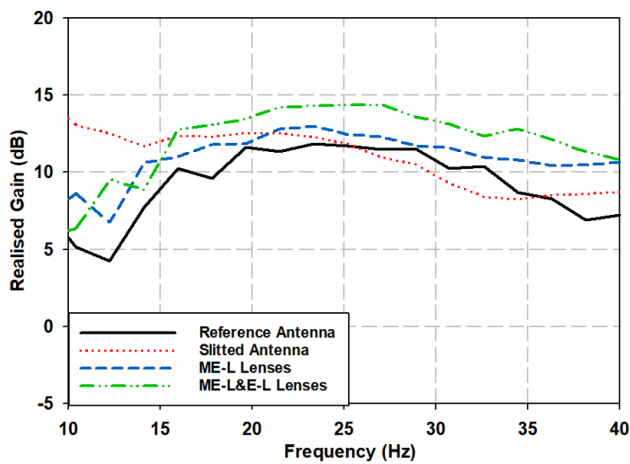


FIGURE 6. Realised gain comparison across different design stages of the proposed BAVA antenna.

gradual improvement in gain. The reference antenna exhibits gain values ranging from 5 dB to 10.5 dB, with its maximum occurring between 20 GHz and 30 GHz. The introduction of slits improves the gain at lower frequencies (10–20 GHz), although a slight degradation is observed beyond 30 GHz compared with the reference design. Incorporating the ME-L lenses yields an additional enhancement of approximately 1.5 dB across the entire band. Furthermore, the inclusion of the E-L lenses pair provides an extra ≈ 2.5 dB improvement relative to the antenna employing only horizontal lenses, particularly within the operational frequency range of interest (20–30 GHz).

C. ANTENNA EFFICIENCY

The total efficiency demonstrates a clear and progressive improvement across the successive antenna design stages, as illustrated in Fig. 7. The reference antenna exhibits the lowest efficiency, varying approximately between 70% and 85% over the operating band, with a noticeable degradation at

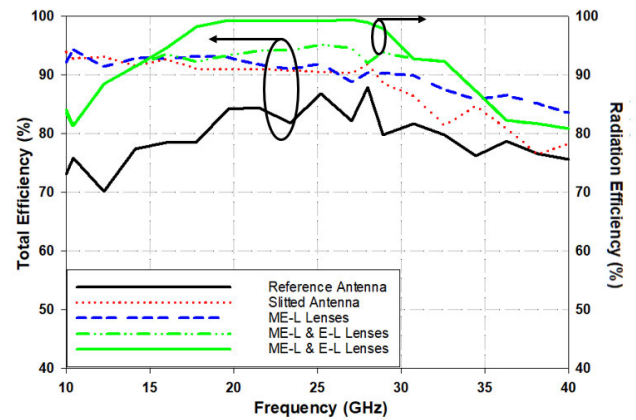


FIGURE 7. Antenna's efficiency comparison across different design stages of the proposed BAVA antenna.

lower frequencies. Following the introduction of open-ended slits, the efficiency improves substantially, remaining predominantly within the 88%–93% range. This enhancement is attributed to the improved impedance transformation between the feed line and the radiating flare, which strengthens impedance matching, reduces reflection and dielectric losses, and promotes a more favourable current distribution.

The addition of horizontal lenses further stabilizes the performance, maintaining efficiencies of about 90–93% across a wide frequency span. The highest performance is achieved when both ME-L and E-L are integrated, where the efficiency exceeds 90% over much of the band and reaches a peak of approximately 94–95% around the mid-band frequencies. In addition, Fig. 7 illustrates the radiation efficiency of the final design, i.e., when both the ME-L and E-L lenses are integrated. It can be observed that, over the frequency range from 15 to 30 GHz, the radiation efficiency reaches approximately 98%, which is about 5% higher than the total efficiency. This behaviour is expected, as the total efficiency accounts for both radiation and mismatch losses, whereas the radiation efficiency reflects only the internal losses within the antenna structure. Despite a slight reduction at the upper end of the spectrum, the combined lenses configuration consistently outperforms all earlier stages, demonstrating its effectiveness in enhancing radiation efficiency and overall antenna performance.

D. RADIATION PATTERN

The normalised radiation patterns in the ϕ -plane at $\theta = 90^\circ$ demonstrate a high degree of stability across the investigated frequency range, as shown in Fig. 8. The main lobe consistently remains directed toward $\phi = 90^\circ$ at all operating frequencies, indicating stable beam pointing with no noticeable beam squint as the frequency increases. In contrast, the radiation pattern becomes progressively narrower with increasing frequency. Specifically, the half-power beamwidth (HPBW) at 10, 18, 28, and 38 GHz is 35° , 23° , 21° , and 16° , respectively. The sidelobe levels remain generally consistent

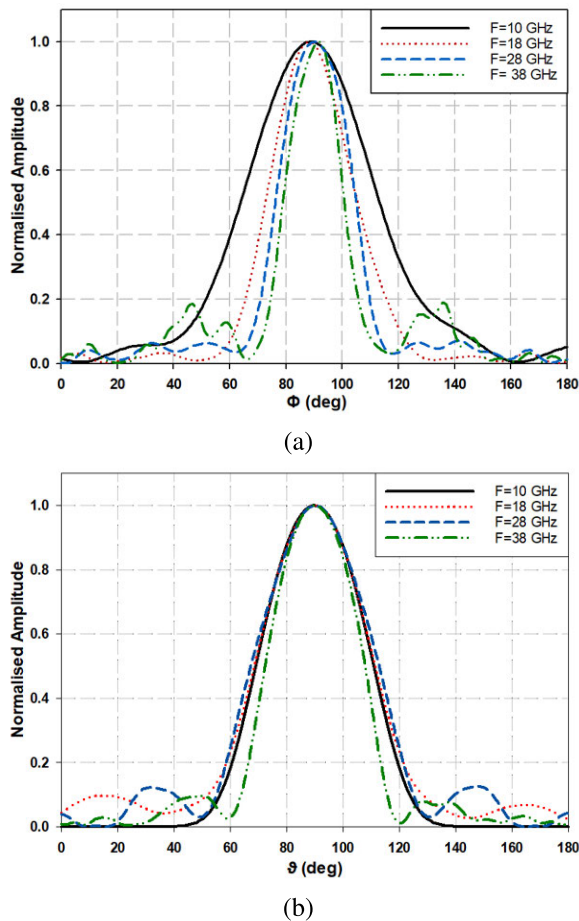


FIGURE 8. Radiation patterns of the proposed AVA antenna at various operating frequencies: (a) azimuthal plane (ϕ -plane) and (b) elevation plane (θ -plane).

across the band, reflecting good pattern control. However, at 38 GHz a slight increase in sidelobe level is observed, reaching approximately 0.2 on the normalised linear power scale, which corresponds to about -7 dB. Despite this minor variation, the overall radiation behaviour remains well preserved, highlighting the frequency-independent performance of the proposed design in the ϕ -plane.

The radiation patterns in the θ -plane at $\phi = 90^\circ$ clearly confirm the end-fire radiation across the operating frequency range, as shown in Fig. 8(b), with excellent beam stability. The antenna exhibits a largely frequency-stable radiation response, with similar pattern shapes and HPBW maintained over most of the band, approximately 30° . Overall, these results further validate the robust end-fire radiation performance of the proposed antenna in the θ -plane.

E. BEAM SQUINT

Fig. 9 illustrates the beam squint at various design stages across the entire frequency range. It can be observed that the reference antenna exhibits a beam squint of approximately 40° within the 10–12 GHz band, and a similar deviation

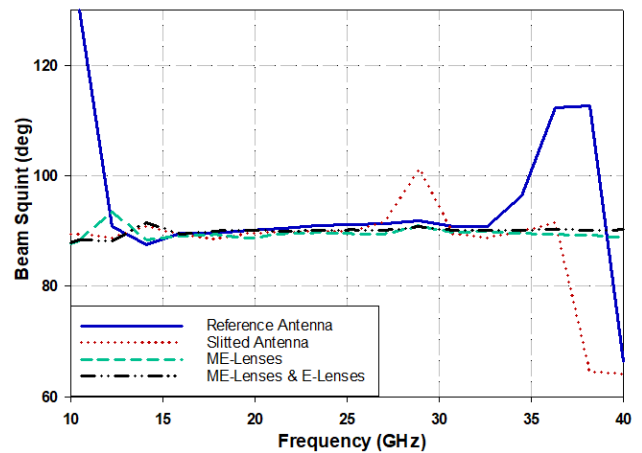


FIGURE 9. Beam Squint comparison across different design stages of the proposed BAVA antenna.

of about 40° reappears at higher frequencies between 32 GHz and 40 GHz. When slits are introduced, the beam deviation within 10–13 GHz is effectively mitigated; however, a noticeable squint persists at frequencies above 35 GHz, along with the emergence of a new deviation between 27 GHz and 30 GHz. Finally, the integration of ME-lenses and E-lenses results in a significant improvement in beam stability, particularly at higher frequencies (from 24 GHz to 40 GHz), where the beam deviation becomes minimal—not exceeding 2° across the entire band.

To further investigate the stability of the radiation characteristics of the proposed antenna, the phase center behaviour has been evaluated. The extracted sigma values, representing the standard deviation of the phase error relative to the best-fit spherical wave, are found to lie within the range of 5.32–5.4 across the extreme ends of the operating band (10–40 GHz). These relatively low values indicate a well-defined phase center, suggesting that the radiated field can be effectively approximated by a spherical wave originating from a single point. This behaviour confirms that the proposed antenna maintains good phase center stability over the considered frequency range.

F. SIDE LOBE LEVEL

In Fig. 10, the sidelobe levels (SLL) of different design configurations are depicted across the entire frequency spectrum. Based on the figure, it is evident that each design stage influences the SLL. The reference antenna exhibits an SLL ranging from -5 to -8 dB within the frequency band of interest (20–30 GHz). It is also observed that introducing slots increases, more precisely, degrades the SLL by approximately 2 dB. In contrast, incorporating ME-lenses significantly reduces the SLL to below -10 dB within the targeted frequency band, and to even lower values at the lower end of the spectrum. The addition of E-lenses, however, does not substantially affect the SLL, as its value remains close to that achieved with the ME-lenses alone. It can be

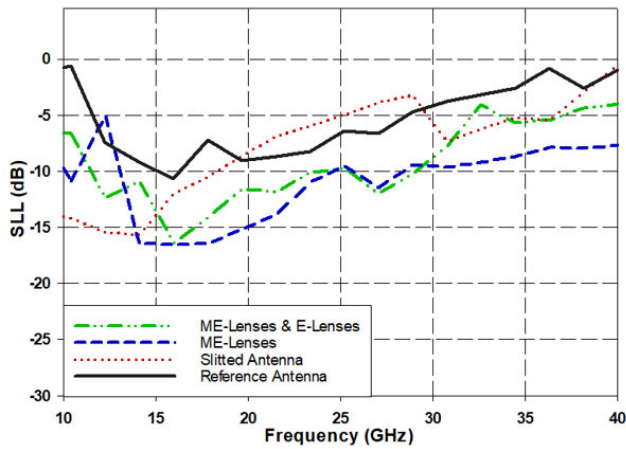


FIGURE 10. Sidelobe level comparison across different design stages of the proposed BAVA antenna.

observed that the SLL degrades beyond 30 GHz for all design iterations. Nevertheless, the implemented modification stages effectively suppress the SLL, maintaining it below -5 dB across the operating band up to 40 GHz.

G. FRONT TO BACK RATIO AND HPBW

Fig. 11 illustrates the front-to-back (F/B) ratio for the different design configurations. It is evident that the reference antenna exhibits an F/B ratio below 20 dB. However, the introduction of slits, followed by the incorporation of ME-lenses and E-lenses, results in a noticeable enhancement in the F/B ratio, particularly within the 15–35 GHz range.

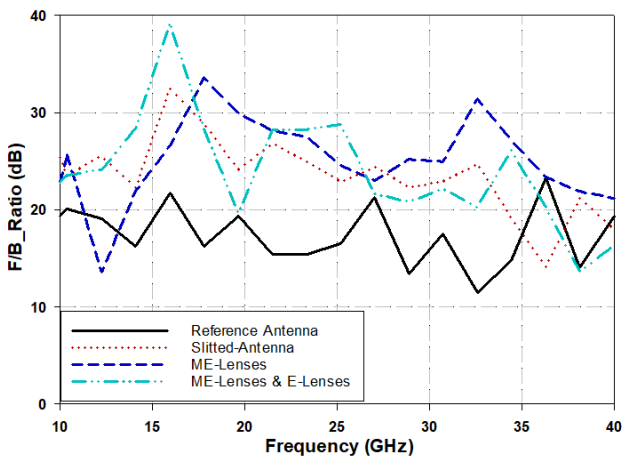


FIGURE 11. Front to back ratio comparison across different design stages of the proposed BAVA antenna.

Fig. 12 further presents the half-power beamwidth (HPBW) of the proposed antenna. A significant instability in HPBW is observed between 10 and 15 GHz for the reference antenna configuration, the slitted antenna, and the design employing ME-lenses. Additional instability is also noted at higher frequencies, especially for the slitted configuration, where the HPBW reaches approximately 80° within the

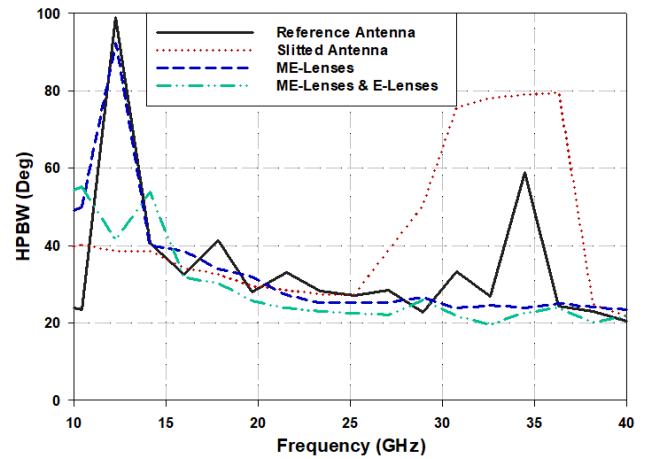


FIGURE 12. HPBW comparison across different design stages of the proposed BAVA antenna.

25–37 GHz band. In contrast, combining both the ME-L and E-L lenses leads to a clear reduction and improved stability of the HPBW across the 20–40 GHz frequency range.

H. FABRICATION TOLERANCE ANALYSIS

This section addresses the sensitivity of the proposed design to fabrication tolerances. In practice, deviations may occur during fabrication, including variations in the positioning of the lenses as well as in the dielectric properties of the substrate. According to the datasheet of Rogers RO4003C, the dielectric constant may vary by approximately $\pm 5\%$ under different operating conditions and frequencies [20].

Accordingly, a tolerance analysis has been carried out to evaluate the robustness of the proposed antenna. Specifically, the positions of the lenses were varied by ± 0.2 mm (left and right), and the dielectric properties were adjusted within the specified ranges. The corresponding impact on the antenna return loss across the frequency band of interest is presented in Fig. 13 (a) and (b) respectively.

The results indicate that the influence of these tolerances is negligible, with no significant variation observed in the return loss. This confirms that the proposed antenna design exhibits high robustness and stability under practical fabrication variations.

IV. FABRICATION AND VERIFICATION

The proposed antenna was fabricated to compare the measured results with those obtained from simulations. The proposed designs were manufactured by Cambridge Circuits Company LTD [21], using Rogers RO4003C as the substrate material, which has a relative permittivity of 3.55, a loss tangent of 0.0027, and a thickness of 0.506 mm, as illustrated in Fig. 14. The overall dimensions of the fabricated single antenna are $64.2 \times 40.3 \times 0.506$ mm³.

The fabricated antenna was characterised at the UKRI National Millimetre Wave Facility using an N5245B vector network analyser (VNA) equipped with 50 Ω 2.4 mm

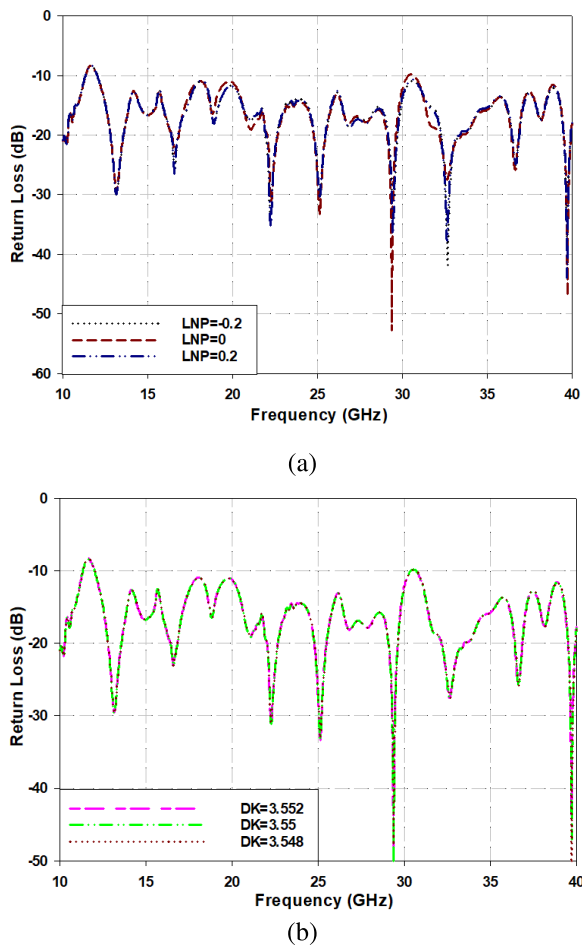


FIGURE 13. Antenna return loss response with respect to variations in: (a) lens position (LNP) and (b) substrate dielectric constant (DK).



FIGURE 14. Photograph of the fabricated prototype of the proposed antenna.

SMA connectors and the NSI-MI Technologies antenna measurement system, as shown in Fig. 15. To ensure measurement accuracy, a full two-port VNA calibration was

performed prior to testing. The antenna under test was mounted on a low-permittivity support inside an anechoic chamber and measured over the operating frequency range under far-field conditions, with radiation characteristics obtained via antenna rotation. Measurement uncertainties may arise from fabrication tolerances, connector and cable losses, and minor alignment errors. Additionally, variations in the feeding structure and probe characteristics at higher frequencies may introduce slight discrepancies between simulated and measured results. Anechoic chambers provide a controlled free-space environment by suppressing reflections [25], making them suitable for accurate radiation pattern characterization, as adopted in this work.

Fig. 16 presents a comparison between the simulated and measured input impedance with a close agreement between simulation and measurement. A slight discrepancy is observed at frequencies above 30 GHz, which can be attributed to fabrication tolerances as well as the operational characteristics of the 2.4 mm RF feedline probe. Nevertheless, the proposed antenna maintains satisfactory impedance matching within the intended frequency band of interest.

Fig. 17 presents a comparison between the normalised simulated and measured radiation patterns in the azimuthal plane, demonstrating excellent agreement over the frequency range from 10 GHz to 38 GHz. Furthermore, the radiation patterns in the ϕ -plane at $\theta = 90^\circ$ confirm that the proposed antenna exhibits a high degree of pattern stability across the investigated frequency band. It is also observed that the radiation beam becomes progressively narrower as the operating frequency increases, while the main lobe direction remains fixed at $\phi = 90^\circ$.

Fig. 17(b) illustrates the comparison between the simulated and measured radiation patterns in the elevation plane. As observed, the radiation pattern maintains a high level of stability with respect to frequency variation, where the beamwidth remains nearly constant across all operating frequencies. Moreover, the main lobe direction is consistently maintained at $\theta = 90^\circ$, indicating that the proposed antenna preserves end-fire radiation characteristics over the entire investigated frequency range.

The antenna gain was measured based on the Friis transmission method [22]. In this measurement, the probe tip of the NSI-MI Technologies spherical measurement system was employed as the receiving antenna, while the antenna under test (AUT) was used as the transmitting antenna. For comparison purposes, a reference antenna provided by the measurement system was utilised, namely the LB-45500-2.4F broadband horn antenna, operating over the frequency range of 4.5–50 GHz with a nominal gain of 13 dB and a 2.4 mm female connector. The gain results are illustrated in Fig. 18, showing a slight discrepancy of approximately 1 dB between the simulated and measured results.

At lower frequencies (10–15 GHz), the measured gain is slightly higher than the simulated gain. However, within the frequency range of 15–30 GHz, the simulated gain exceeds the measured gain by approximately 1 dB. A similar trend

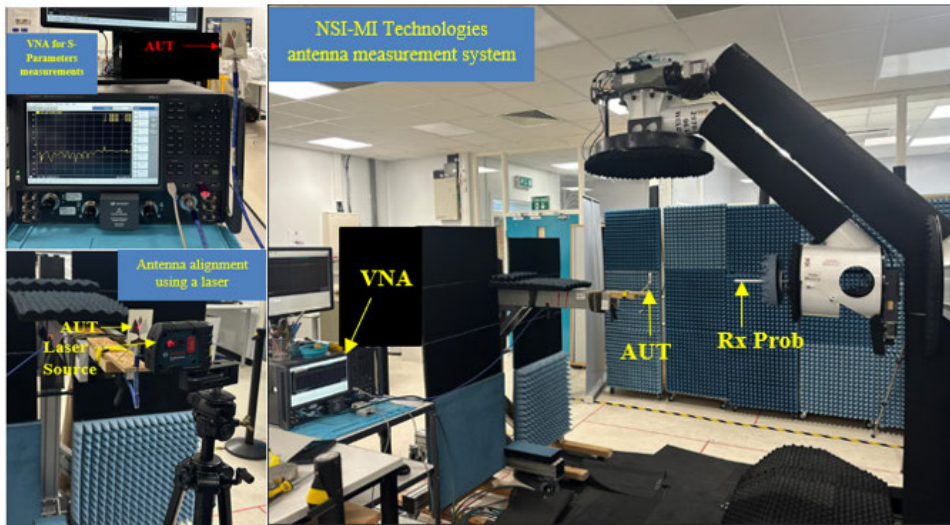


FIGURE 15. Practical measurement system setup.

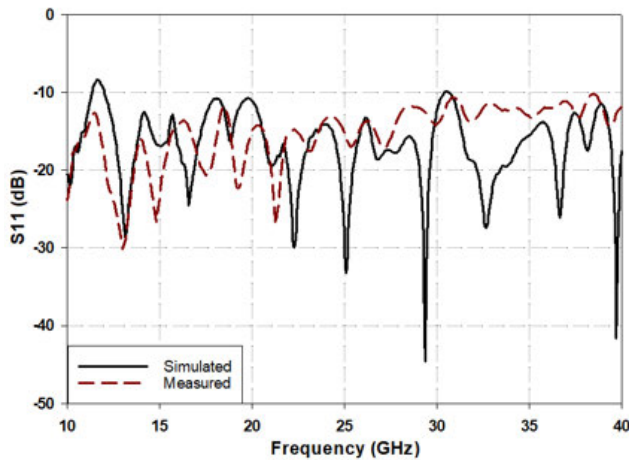
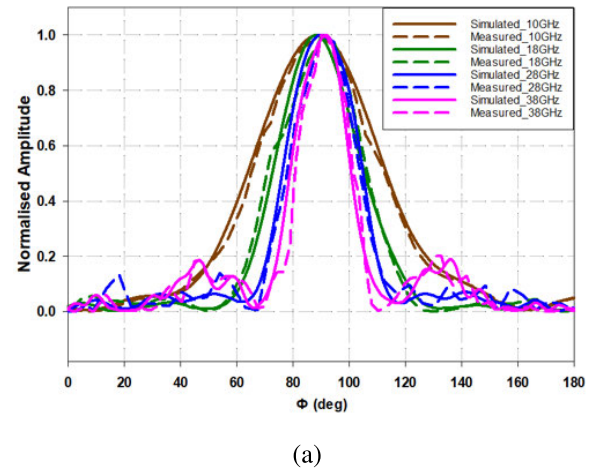


FIGURE 16. Simulated and measured return loss of the proposed Vivaldi antenna.

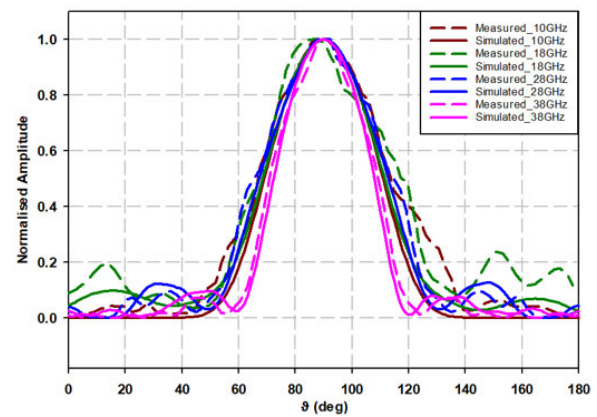
is observed at higher frequencies between 30 GHz and 40 GHz.

These discrepancies can be attributed to several factors, including fabrication tolerances, substrate material uncertainties, measurement setup misalignment, and idealised assumptions inherent in the simulation models. Nevertheless, it is noteworthy that the measured gain exhibits better stability compared to the simulated results. Moreover, both simulated and measured gains demonstrate a gradual increase as the operating frequency increases.

To highlight the merits of the proposed antenna in comparison with previously reported designs, Table 1 presents a comprehensive comparison between the present work and earlier studies. It can be observed that most designs achieving relatively high gain, such as [7], [8], [9], and [23], rely on dielectric lenses. This technique is characterised by increased fabrication and alignment



(a)



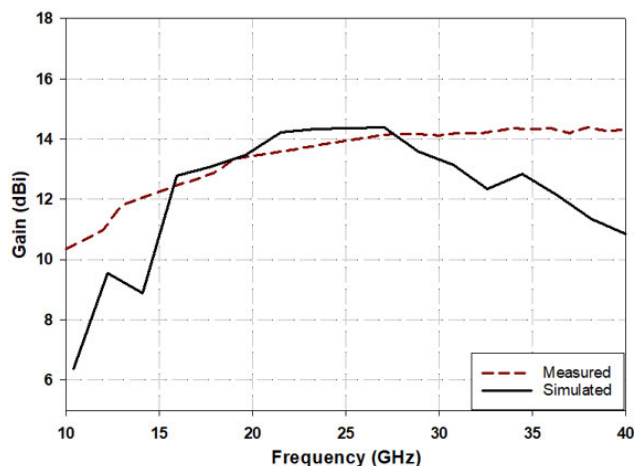
(b)

FIGURE 17. Radiation patterns of the proposed AVA antenna at various operating frequencies: (a) azimuthal plane (ϕ -plane) and (b) elevation plane (θ -plane).

complexity, as the carving and shaping of a dielectric lens, whether with a permittivity different from or identical to

TABLE 1. Performance comparison between the proposed antenna and previously reported designs.

Reference	Band (GHz)	Size ($\lambda \times \lambda$)	Enhancement	Gain (dB)	Δ Gain (20–40 GHz)	Beam Squint
[23]	10–40	$2.66\lambda \times 0.80\lambda$	Substrate cutout + slotted edges	9.6–15	≈ 5.4	≤ 3
[7]	9–40	$2.33\lambda \times 1.07\lambda$ + Lens $\approx 1.07\lambda$	Half-spherical dielectric lens + parasitic patch	13.18–20.54 (Simulated)	≈ 7.5	2
[24]	3.7–18	$0.91\lambda \times 0.78\lambda$	None	1.8–6.9	N/A	N/R
[8]	3.4–40	$0.95\lambda \times 1.42\lambda$	Trapezoid-shaped dielectric lens	Up to 14	≈ 4 –5	≤ 3
[10]	1–28	$0.63\lambda \times 0.29\lambda$	Director + metamaterial (no lens)	11.6–14.6	N/A	N/R
[26]	2–32	$0.79\lambda \times 0.37\lambda$	Parasitic elliptical patch (no lens)	>10 (6–21 GHz), ~ 0 (21–32 GHz)	N/A	Not quantified
[9]	4.2–42	$1.63\lambda \times 0.58\lambda$	Koch fractal parasitic lens + fractal dielectric lens	0–13	≈ 4 –5	Not quantified
[5]	6–26.5	$1.34\lambda \times 0.71\lambda$	Double asymmetric trapezoidal parasitic patch	>9	N/A	Not quantified
[12]	0.98–35	$0.28\lambda \times 0.29\lambda$	None	~ 6 –10 (array directivity)	Not applicable	Not quantified
This Work	10–40	$2.14\lambda \times 1.34\lambda$	ME-L & E-L Lenses	10.3–14.3	≤ 1	≤ 2

**FIGURE 18.** Simulated and measured return loss of the proposed Vivaldi antenna.

that of the antenna substrate, and its subsequent placement within the aperture slot or attachment to the antenna body significantly increase mechanical complexity and sensitivity to manufacturing and positioning tolerances. Furthermore, such configurations introduce additional dielectric losses and perturb the near-field distribution around the aperture, which may adversely affect gain stability and radiation characteristics, particularly at millimetre-wave frequencies.

On the other hand, designs that do not employ dielectric lenses generally exhibit narrow operating bandwidths and relatively low gain, as reported in [5], [10], [12], [24], and [26]. In addition, some of these works demonstrate gain instability or do not report gain stability and beam squint characteristics, which may be significant in such configurations. Moreover, several of these designs also suffer from relatively large physical dimensions. In contrast to conventional lens-based approaches, the proposed

antenna employing composite ME-L and E-L lenses provides wideband phase correction through an electrically thin and compact configuration. The antenna achieves a high and stable realised gain of 14.5 dBi with beam squint below 2° across the 20–40 GHz band. The proposed structure enhances gain flatness and beam stability while maintaining a low-profile, planar, and automated-based fabrication structure. Consequently, it avoids the size enlargement, multilayer complexity, and material-integration limitations associated with external dielectric lenses, thereby offering strong suitability for wideband 5G millimetre-wave systems.

V. CONCLUSION

This work has presented a novel antipodal Vivaldi antenna architecture integrated with two complementary lens types—magneto-electric lenses and electric lenses targeting millimetre-wave applications across the 10–40 GHz band. The ME-L structure enables the simultaneous excitation and controlled coupling of electric and magnetic field components, effectively concentrating the near-field distribution toward the aperture. This mechanism enhances aperture efficiency, reduces lateral field leakage, and leads to measurable improvements in realised gain. In parallel, the E-L pair introduces predominantly electric coupling and operates as a phase-correcting director system, redistributing the aperture field to produce a more uniform phase front and strengthened end-fire radiation, with additional gain enhancement and beam narrowing, particularly in the elevation (θ) plane. Furthermore, the incorporation of open-ended elliptical slits contributes to broadband impedance matching, especially at the lower frequencies of the operating range. The combined design approach achieves wideband performance with high and stable gain of 14.5 dBi and significantly reduced beam squint (less than 2°) over the entire 10–40 GHz band. In addition to its electromagnetic performance, the proposed antenna-lens configuration maintains a compact, low-cost

planar profile and can be fabricated as a single integrated structure, thereby supporting practical manufacturability and suitability for advanced mmWave 5G systems. Moreover, it eliminates the need for manual integration of a separate lens, which represents a key practical challenge at millimetre-wave frequencies.

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